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Chapter # 13: Cytokines

Overview about Cytokines

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Cytokines belong to diverse group of non-antibody proteins which act as mediators between cells. Initially, these proteins were considered as products of immune cells. More importantly, these were also considered to act as mediators for immune cells. Later on, it was also found that these are also produced from non-immune cells and served as mediators too. Cytokines are used as biological response modifier for treatment of various diseases. In general, cytokine is a general term but it can be more specific term which is used according to their cell of origin as follows

- Monokines

These are produced by mononuclear phagocytic cells.

- Lymphokines

These are produced by activated lymphocytes.

- Interleukins

These are the mediators between leucocytes

- Chemokines

These are responsible for leucocytes migration and inflammatory process

Cytokines usually perform their functions as cascade signaling in the form of a network which is termed as cytokine network. Their functions are for enhancing their effects of other cytokines in an additive or synergistic manner. Moreover, these act for suppressing the effects of other cytokines in an opposing or antagonistic way. Cytokines are not as preformed proteins like antibodies. These are produced by gene transcription as needed because mRNA for cytokines are short lived in nature. Furthermore, the individual cytokine can act on many cells of the host which is termed as pleotropic response of cytokines.

Mechanism of acting of Cytokines

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Cytokines act through specific receptors present on various cells. These are considered as redundant in nature. The redundancy of cytokines is due to the nature of cytokine receptors. The complex interaction between host's cells and cytokines occur through cytokine network. Different cells can respond to same cytokine due to structural similarities among cytokine receptors (Fig.13.1). The cytokine signaling is flexible which can have both protective and damaging effects. Similarly, these can also influence the synthesis of other cytokines.

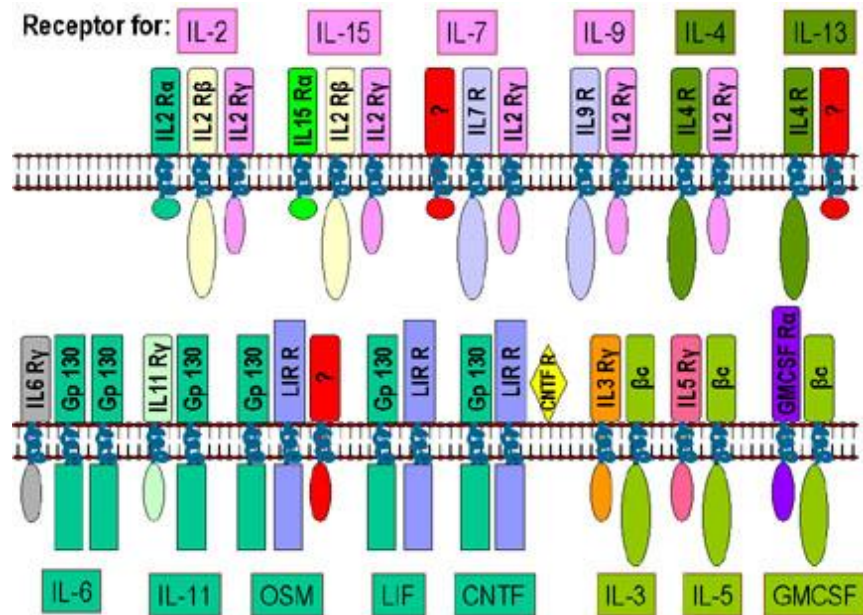


Figure 13.1: The structure of various cytokine receptors

Cytokines bind to specific receptors with high affinity and respond in following three ways

1. Autocrine

These effect on same cells which secrete the cytokine

2. Paracrine

In this way, the cytokines effect on nearby cells of cytokine secreted cells.

3. Endocrine

Through this way, the cytokines effect on distant cells through circulation.

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Categories of Cytokines

Cytokines are categorized on following two basis

I. Their source of origin

II. Their functions

I. Categories based on source cells

Followings are the categories of cytokines on the basis of their source of origin

i) Monokines

As the name indicates, these are produced by mononuclear phagocytic cells for example Interferon gamma and Interleukin (IL-1) from macrophages & Monocytes

ii) Lymphokines

These are produced by activated lymphocytes for example IL-3, IL-4 & IL-5.

iii) Interleukins

These are mediators between leucocytes for example IL-1, IL-10 & IL-18.

iv) Chemokines

These are responsible for leucocytes migration for example IL-8.

II. Based on functions

On the basis of functions, cytokines are categorized into following two categoryies

a) Cytokines which act for Innate (non-Specific) immune system e.g TNF- α , IL-1, IL-10 etc

b) Cytokines which act for Adaptive (Specific) immune system e.g IL-2, IL-4 & IL-5

Cytokines of Innate Immune System

These cytokines which act primarily for innate (non-Specific) immune system. These are as follows

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1) Tumor Necrosis Factor-Alpha (TNF- α)

It is produced by activated macrophages in response to microbes or their products for example lipopolysaccharides (LPS) of gram negative bacteria. It acts a mediator of acute inflammation. It recruits neutrophils and macrophages at the site of infection either by stimulating endothelial cells to secrete adhesion proteins or by secreting chemokines. Moreover, TNF- α induces fever and acute phase proteins in host.

2) IL-1

This is also an inflammatory cytokine as like TNF- α . It is produced by activated macrophages. Furthermore, it also help to activate T-cells.

3) IL-10

It is an inhibitory cytokine which inhibits the cytokine production of cytokines from macrophages.

4) IL-12

This cytokine is produced by dendritic cells and macrophages. It enhances the cytolytic activity of cytotoxic T-cells

5) Type I interferon

This class of interferon include Interferon α & β . These inhibit viral replication in cells by increasing the expression of class I MHC which in turn make them susceptible to cytotoxic T-lymphocytes (CTL).

6) IFN- γ

It is produced primarily by Th1 cells. It enhances the cytolytic activity of NK cells by induction of class I & II MHC molecules.

7) Chemokines

These are chemotactic cytokines produced by many leucocytes. These recruits inflammatory cells at site of infection for an inflammatory response.

Cytokines of Adaptive Immune System

Followings are the cytokines which act primarily for Adaptive (Specific) immune system

1) IL-2

It is produced by T-helper (Th) cells but Cytotoxic T-cells (Tc) also produced it in lesser extent. It stimulates B-cell division. It can also activate NK cells and monocytes. Moreover, IL-2 acts on T-cells in an autocrine manner.

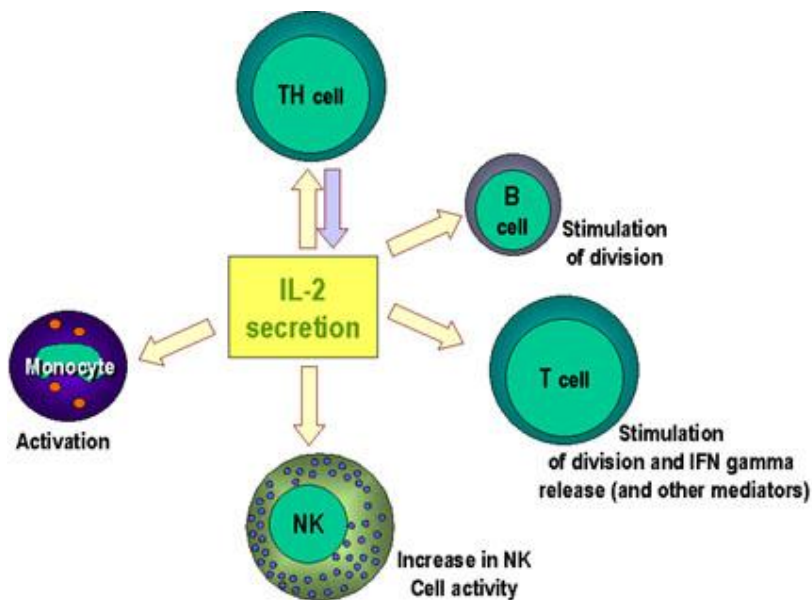


Figure 13.2: Functions of IL-2 on various other immune cells

Activation of T-cells results in expression of IL-2R receptors and also production of IL-2. IL-2 binds to IL-2R and promotes cell division. However, IL-2 receptor decays if T-cells are not activated and eventually the proliferative phase ends (Figure 13.3).

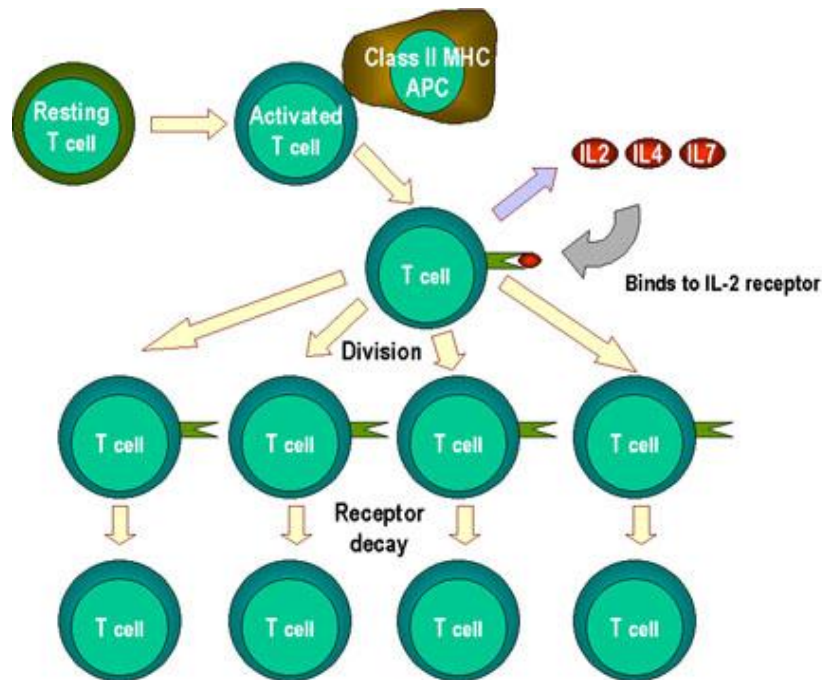


Figure 13.3: Activation of T-cells by IL-2

2) IL-4

IL-4 is produced by macrophages and Th2 cells. It promotes the development of Th2 from naïve T-helper (Th) cells. The Differentiated Th2 cells result in the production of antibodies. IL-4 also involves in the class switching of antibody into IgE

3) IL-5

IL-5 is produced by Th2 cells. It promotes the development of B-cells and eosinophils. It also activates mature eosinophils.

4) Tumor Growth Factor (TGF)- β

TGF- β is produced by T cells and many other cells types. It acts as inhibitory cytokine which inhibits the proliferation of T-cells and activation of macrophages. It also acts on neutrophils and endothelial cells to block the effects of pro-inflammatory cytokines.

Cytokine Network

A complex series of overlapping and inter-related connections among cytokines. As cytokines are secreted from one kind of cells and have effect on other type of cells & organs. Within this network, some cytokines have synergistic effects on other cytokines while some have antagonistic effects. Through this network, lymphocytes, macrophages and other tissues are regulated for their functions (Fig. 13.4)

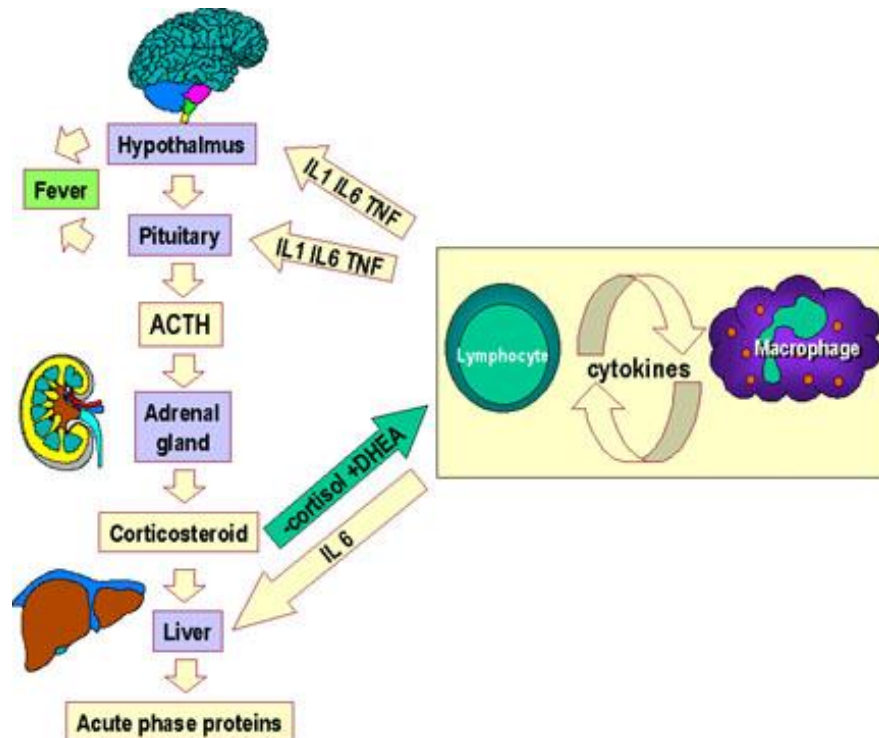


Figure 13.4: Communication B/W Lymphocytes, Macrophages & Other Tissues

Similarly, monocytes, lymphocytes and other components are also communicated through this network (Fig. 13.5)

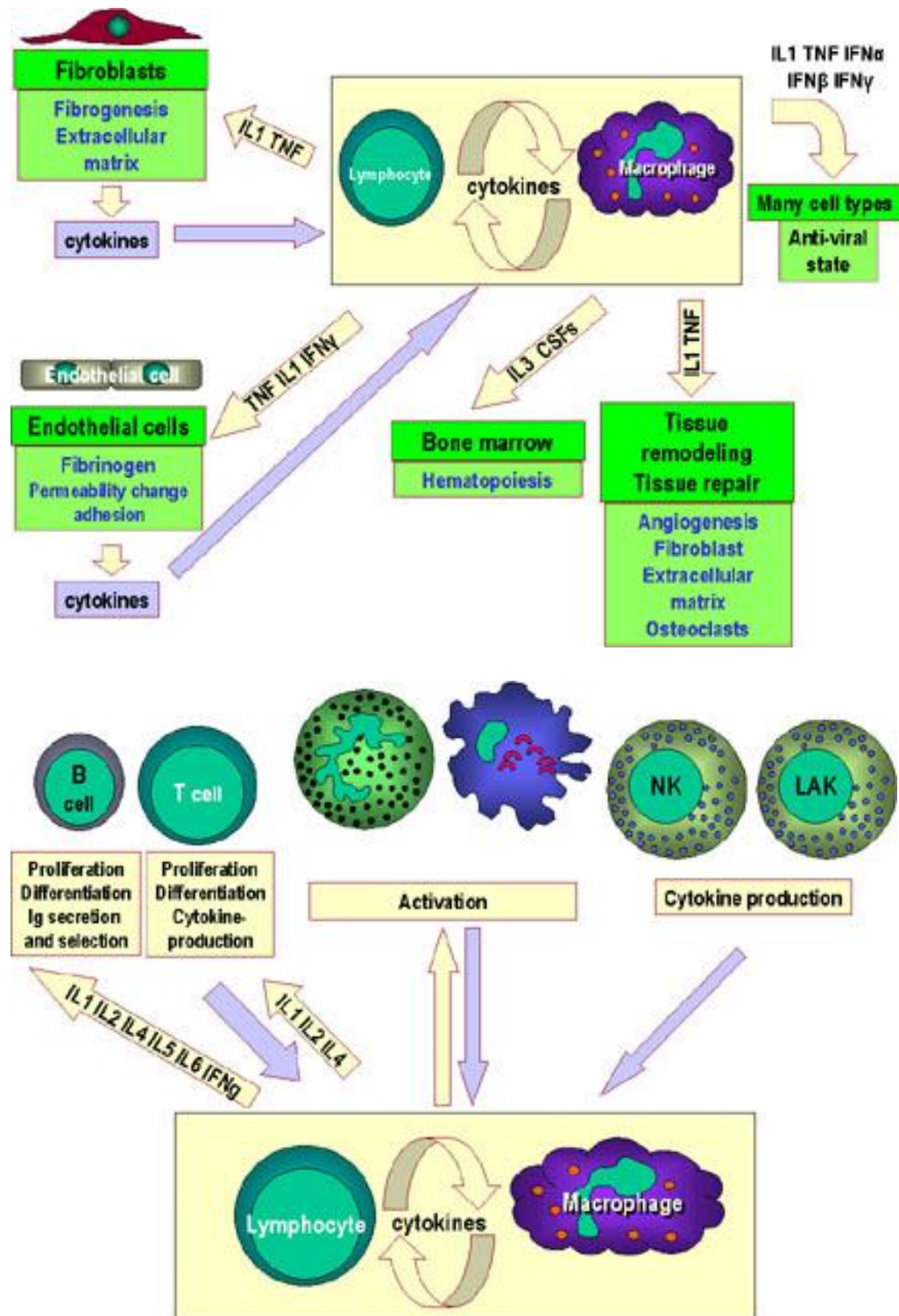


Figure 13.5: Communication B/W Lymphocytes, Macrophages & Other Components of Immune System

Immuno-Regulation by Cytokine

The control of immune response between lymphocytes and macrophages. The balance is required between antigen driven activation of lymphocytes and regulatory influences which is called as immune-regulation. Immuno-regulation occur at following three phases of immune responses

- 1) Recognition phase
- 2) Activation Phase
- 3) Effector Phase

The cytokines are considered as positive or negative regulator of immune response. The cytokines can act on many stages of immune responses. Their activity is dependent on presence of other cytokines in the tissue's microenvironment. It also depends upon receptor expression on effector cells. This kind of regulation regulates the nature and extent of generated immune response.

Chapter#14: Resistance and Immune Response to Infectious diseases

What is Immune Evasion?

All the strategies used by pathogenic organisms and tumor cells to evade host immune's system are called as immune evasion mechanisms. Immune system has one of the important characteristics of immune surveillance for protection against pathogenic organisms and transformed cells. However, pathogens maximize their ability to flourish and develop infection. Likewise, tumor cells also have the potential to evade the host's immune surveillance and sustain inside the body. Immune-evasion occurs either due to weak host immune response or by strategies devised by pathogens. In immune evasion, immune system can be bypassed by evading humoral immunity. Moreover, cell mediated immunity can also be evaded by pathogenic microbes and transformed cells.

Mechanisms of Immune Evasion

The strategy to escape from immune system is caused by different microbes differently as follows

a) Extracellular organisms

These organisms usually inactivate the humoral components of immune system

b) Intracellular organisms

These organisms usually bypass the host immune system by inactivating intracellular killing and other cell mediated immune mechanisms which are as follows

Antigenic Variation

Microorganisms mutate their antigenic surface molecules for immune-evasion which is called as antigenic variation. As a result, there is not available longer protection by antibodies produced under previous exposure of antigenic forms. There are following two forms of antigenic variations

- Antigenic shift

This is the major form of change in antigenic structure of antigen as a result the new strains of microbes are generated (Fig.14.1)

- Antigenic drift

This is the minor form of antigenic variation within the same strain for example influenza (Flu) virus exhibits both antigenic shift and drift for pandemic and seasonal flu respectively. Similarly, bacteria like *E.coli*, *Neisseria gonorrhoeae* and *Salmonella typhimurium* also exhibit antigenic drift (Fig. 14.1).

Differences Between Antigenic shift & Antigenic drift

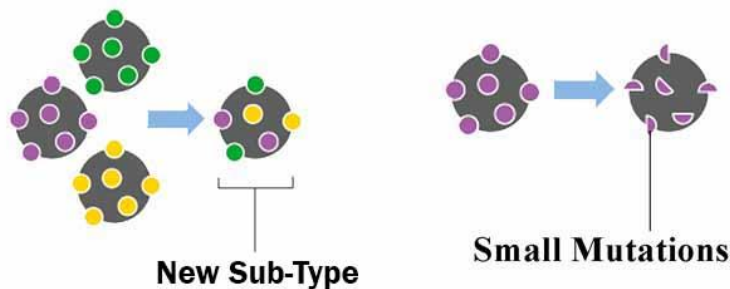


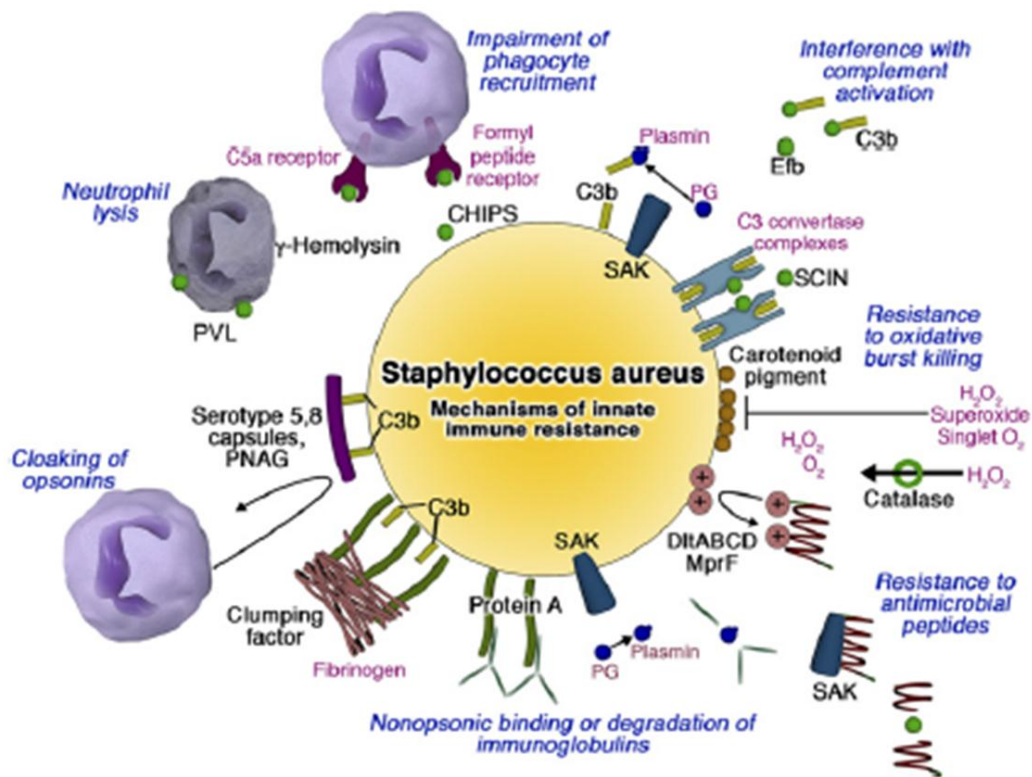
Figure 14.1: Comparison between antigenic shift and drift

Inhibition of Complement activation

The degradation of complement proteins through complement deviation which occurs by deviation of the complement activation site on bacterial cell. Moreover, the resistance to insertion of Membrane Attack Complex (MAC) in bacterial due to thick bacterial cell wall and capsule also responsible for inhibition of complement activation.

Resistance to Phagocytosis

The inhibition of phagocytosis process of bacteria due to cell surface molecules of bacteria for example capsule of bacteria like pneumococcus. Similarly, the trapping of bacteria by bacterial enzymes for example coagulase of *S. aureus*. Moreover, the killing of phagocytes due to various secreted toxins from different bacteria for example Leucocidin & Lysins from *S. aureus* and Streptococci respectively. The different immune-evasion mechanisms of *S. aureus* (Fig.14.2)

Figure 14.2: Strategies of Immune evasion by *S. aureus*

Immune Response against Extracellular Pathogens

The secretory or humoral immune molecules are effective against extracellular pathogens. Various secretory molecules like antibodies and complement proteins contribute in such kind of protection

There are three ways of controlling

- 1) Neutralization
- 2) Opsonization
- 3) Complement activation
- 1) Neutralization

The pathogenicity of bacteria is being determined by secreted molecules of bacteria like toxins. These toxins are neutralized by specific antibodies against toxins in a form of antitoxins by the process of neutralization. The binding of antibodies with toxins leads towards immune complex formation. Furthermore, the clearance of immune complex is done by the process of phagocytosis.

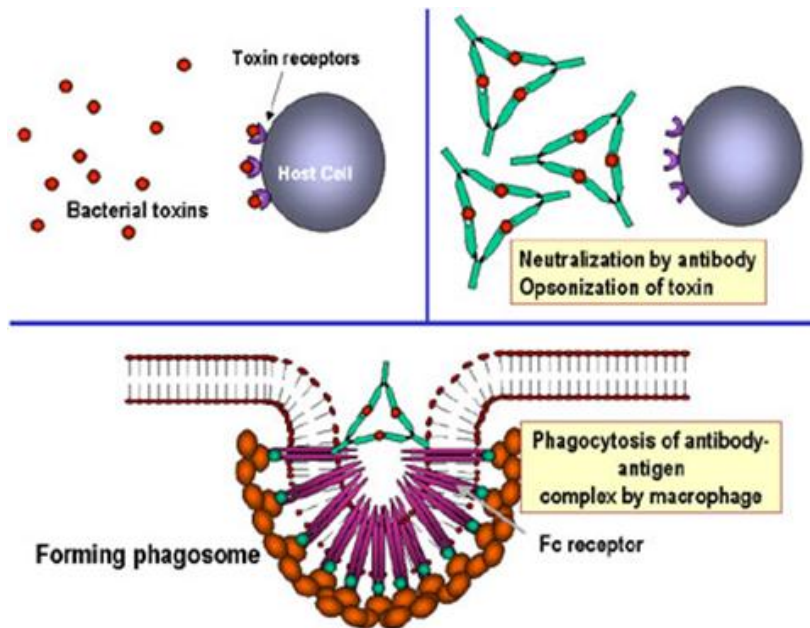


Figure 14.3: The process of neutralization

2) Opsonization

The enhancement of the process of phagocytosis by phagocytes is called as opsonization. The antibodies which bind specifically with extracellular bacteria e.g IgG or IgM are called as opsonins. This result in the formation of immune complex. Moreover, the binding of immune complex occur with specific receptors against Fc fragment of antibodies (Fig. 14.4).

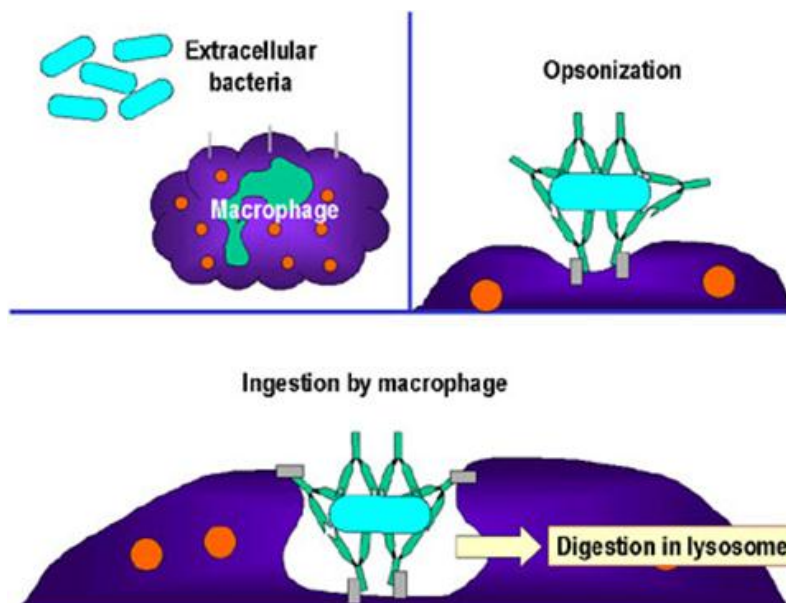


Figure 14.4: The process of opsonization

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Complement activation

The inactive complement proteins are activated by extracellular bacteria in combination with specific antibody. Further, the complement proteins are fixed on antibody which is already bounded with bacterial antigen on bacterial surface. This cause the lysis of bacteria which is the primary function of complement proteins by forming membrane attack complex (MAC) (Fig. 14.5).

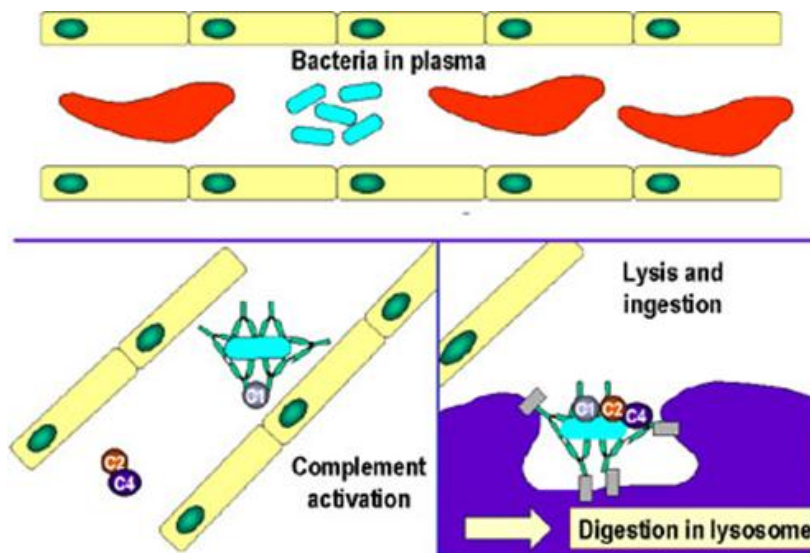


Figure 14.5: The process of complement activation

Immune Response against Intracellular Pathogens

The secretory immune molecules are ineffective against intracellular pathogens like viruses and intracellular bacteria. The cell mediated immune response is the primary defense against intracellular pathogens. The T-Lymphocytes play vital role in cell mediated immunity. Cell mediated immune response varies according to the residing site of the pathogen

- 1) Cytosolic site
- 2) Vesicular site

Control of Cytosolic Pathogens

The exogenous pathogens are the pathogens which used to reside in the cytosol of infected cell. Their antigens are presented in combination with class I MHC to Cytotoxic T-Lymphocytes: which recognize such antigens .Viruses are controlled through such mechanism (Fig 14.6)

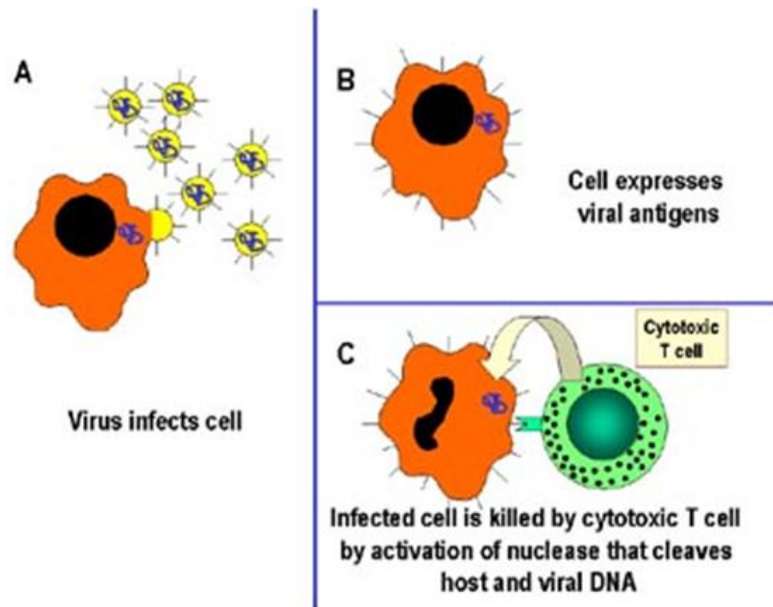


Figure 14.6: The control of cytosolic pathogens

Control of Vesicular Pathogens

The endogenous pathogens are those pathogens which used to reside in the vesicle like phagosomes of infected cell. The antigens of such pathogens are presented in combination with class II MHC to Helper T-Lymphocytes which recognize such antigens (Fig. 14.7). Intracellular bacteria e.g *Mycobacterium*, *Listeria*

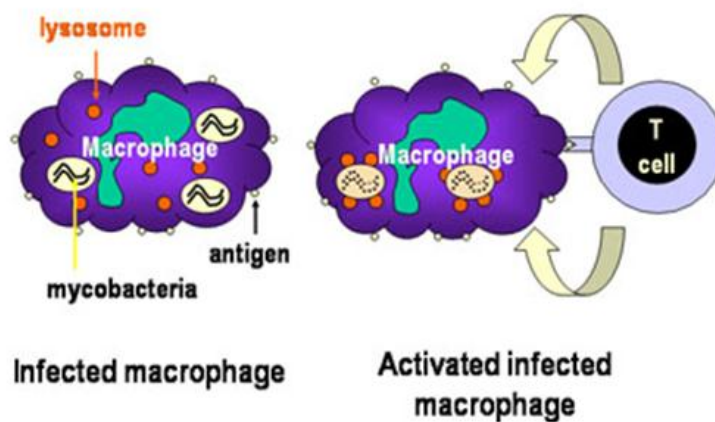


Figure 14.7: The control of vesicular pathogens

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Followings are the ways by which intracellular bacteria evade the host immune mechanisms (Table 14.1)

Mechanisms of Immune Evasion by intracellular Bacteria

Mechanism of Immune Evasion	Examples
Inhibition of phagolysosome formation	<i>Mycobacterium tuberculosis</i> , <i>Legionella pneumophila</i>
Inactivation of reactive oxygen and nitrogen species	<i>Mycobacterium leprae</i> (phenolic glycolipid)
Disruption of phagosome membrane, escape into cytoplasm	<i>Listeria monocytogenes</i> (hemolysin protein)

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Table 14.1: Mechanisms of Immune Evasion by Intracellular Bacteria

Chapter # 15: Cell Mediated Effector Response

Cell Mediated Immunity

The immune response which is independent of antibodies and based on cell to cell interaction is called as cell mediated immune response or immunity. In this kind of immunity, there is activation of phagocytes and antigen specific cytotoxic T-lymphocytes which further release various cytokines from activated cells in response to various antigen. Moreover, naïve & mature T-cells are differentiated into effector T-cells after interacting with antigens via antigen presenting cells (APC). On contrary, the humoral immunity is based on cell free body fluid like serum which contain various protective molecules like antibodies. However, in cellular immunity is associated with cells for protective functions of immunization. Both innate and adaptive immune systems have humoral & cellular immunity

Cellular mediated immunity provides protection by following ways

- T-cell immunity

In this form of protection, cytotoxic T-cells (CTL) are activated for the killing of infected and transformed cells

- By activating macrophages

By this way, macrophages are activated for destroying various pathogens

- By activating natural killer (NK) cells

In this way of protection, NK cells are activated for killing of transformed and viral infected cells

Role of Cytotoxic T-Lymphocytes in Cell Mediated Immunity

The cytotoxic T-Lymphocytes are also called as CTLs and CD8+ve. These cells are not mature after exiting thymus where these have functional TCR which can recognize antigens but cannot kill the target cells. There is needs of differentiation for fully activation of CTLs. CTL differentiates from Pre-CTLs in response to following two signals

1. The specific antigen in context of Class I MHC
2. The cytokines produced from T-helper 1 (Th1) cells especially IL-2 & INF- γ (Fig.15.1)

CTLs cause the killing of target cells. This of killing is antigen specific where target cell must bear same antigen with class I MHC molecule

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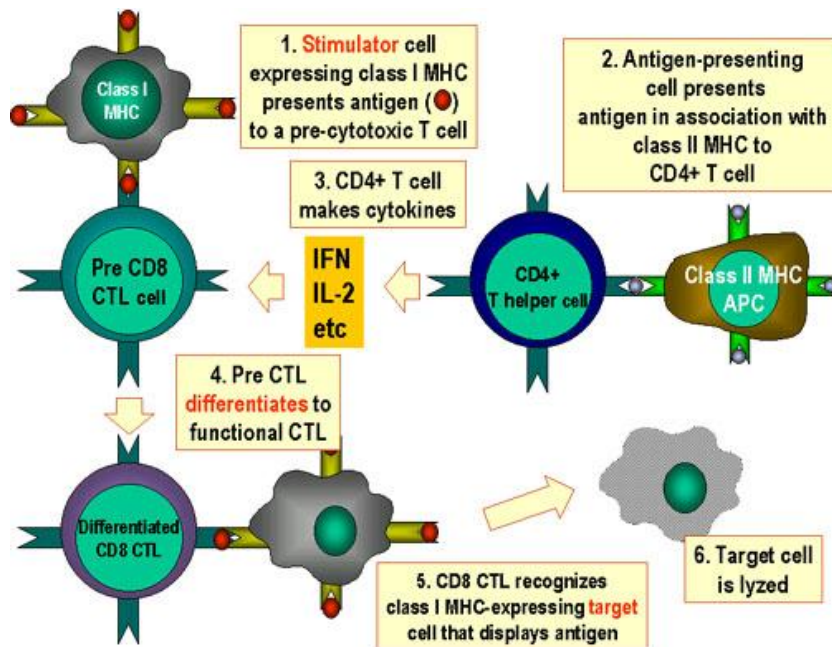


Figure 15.1: Maturation & Differentiation of cytotoxic T-cells (CTLs)

Moreover, CTL killing also requires cell to cell contact in which target cell should contain cell surface MHC molecule. However, CTLs are not injured while targeting the target cell. One CTL can kill sequentially numerous target cells.

Mechanisms of CTL killing

Mechanistically, CTLs rely on granule-mediated killing. In this form of killing, fully differentiated CTL have numerous granules which contain perforin & granzymes in its cytoplasm. Upon CTL degranulation, perforin monomers are polymerized to produce polyperforin molecules. These polyperforin molecules are assembled in the form of channels in the presence of calcium ions (Fig.15.2)

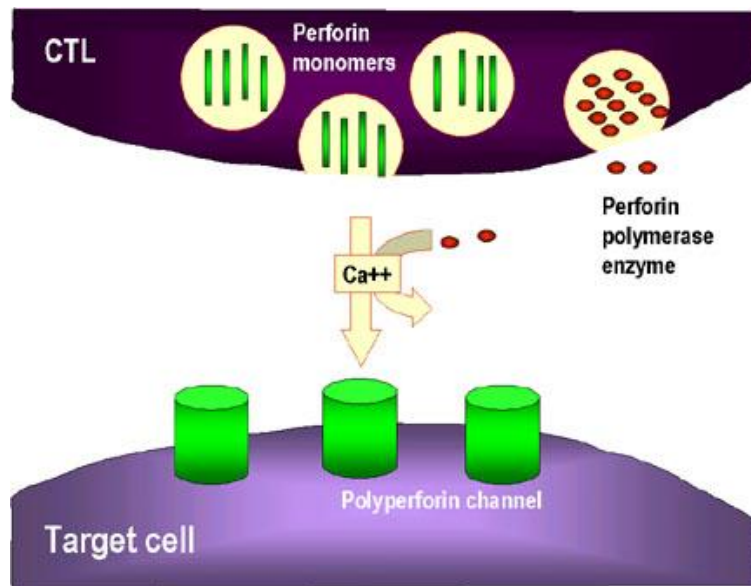


Figure 15.2: Release of perforin molecules from CTL and formation of polyperforin channel in the membrane of target cell

After developing polyperforin channels in the membrane of target cell, the CTL release degradative enzyme or toxins which travel through perforin channels towards target cells (Fig. 15.3). The killing of target cell occurs with the help of degradative enzymes. Furthermore, the cytokines like $\text{TNF-}\alpha$ & $\text{INF-}\gamma$ from CTL bind to target cells and induce programmed cell death or apoptosis of target cell

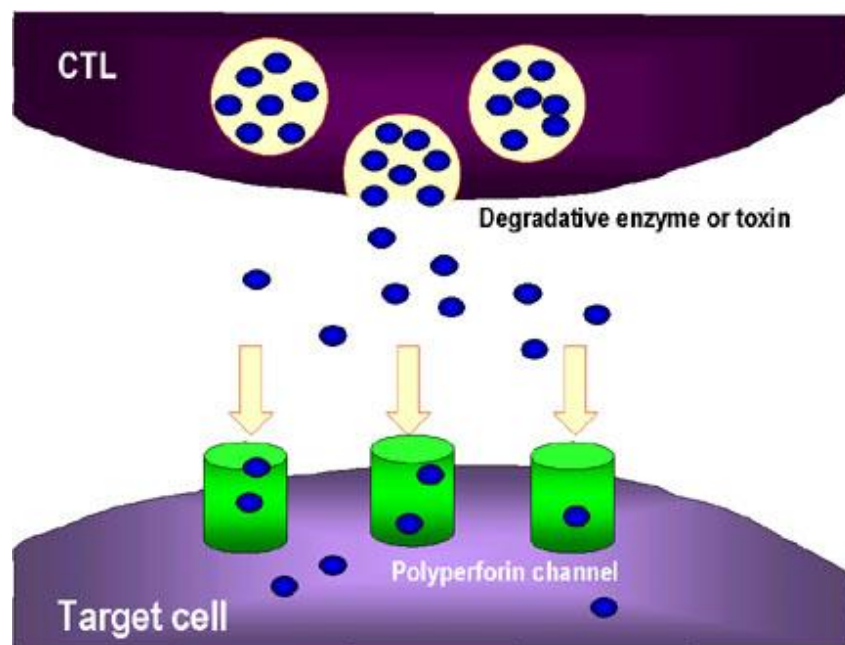


Figure 15.3: Killing of target cell after releasing of degradative enzyme or toxin

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Role of Helper T-Lymphocytes in Cell-Mediated Immunity

Helper T (Th)-Lymphocytes which are also called as CD4+ve cells. These cells are at the center of cell mediated immunity. Helper T-lymphocytes recognize specific antigens presented with class II MHC. Moreover, these cells are also involved in B-lymphocytes activation. Helper T-cells also release various cytokines which further activate other immune cells (Fig.15.4)

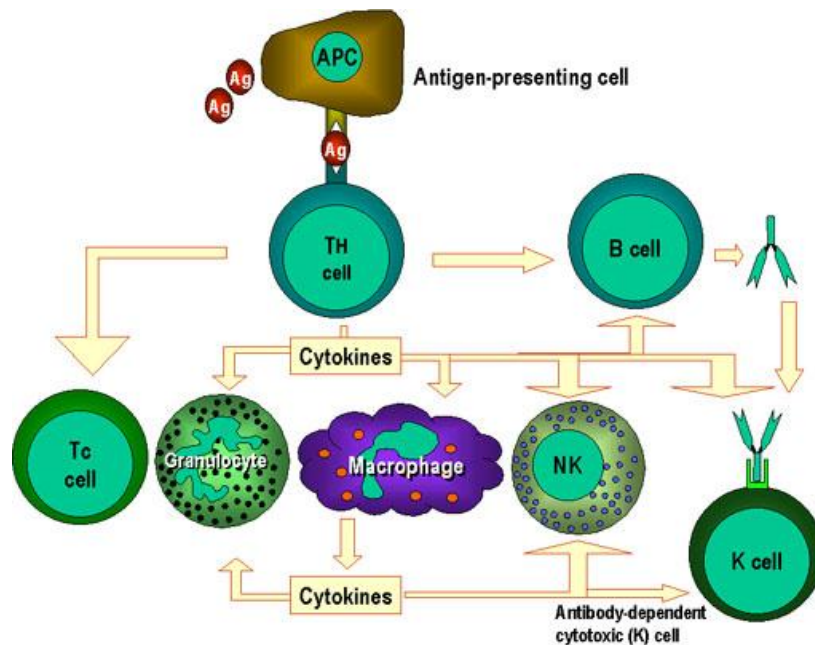


Figure 15.4: Functions of Helper T-lymphocytes

There are two major subtypes of Th cells

- 1) Th1 cells
 - 2) Th2 cells
- Th1 cells

These cells release cytokine like $\text{IFN}\gamma$ upon activation which then activate macrophages. In turn, macrophages participate in the generation & differentiation of CTL for cell mediated immunity. However, $\text{IFN}\gamma$ acts as inhibitory cytokine for Th2 (Fig.15.5)

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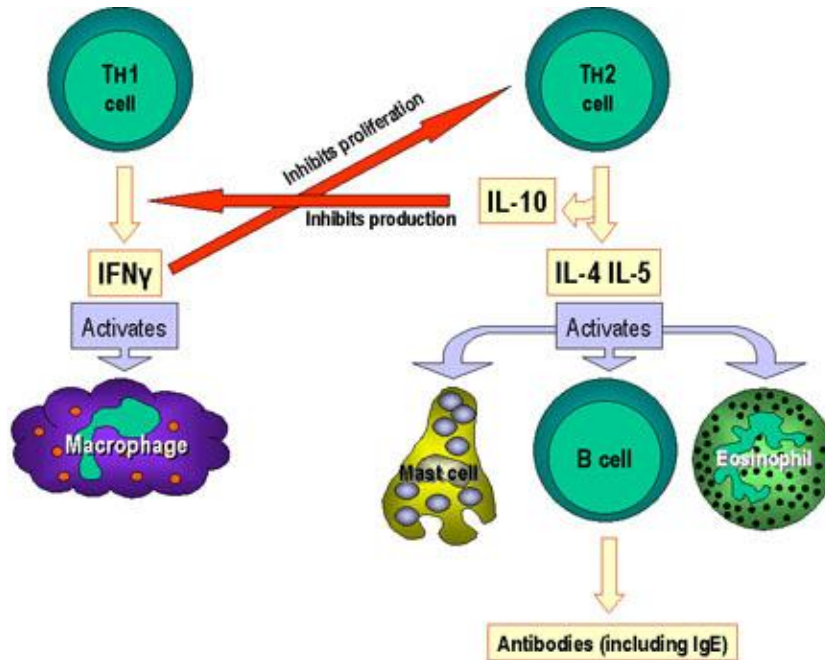


Figure 15.5: Function of Th1 & Th2 cells

- Th2 cells

Upon activation, Th2 cells secrete cytokines like IL-4 & IL-5 which further activate B-cells for humoral immunity. This kind of humoral response is also involved in the generation of IgE antibodies through the process of class switching. Moreover, Th2 cells are also involved in the activation of mast cells and eosinophils for providing immunity against allergen and parasitic infections. Th2 cell also secrete IL-10 which is inhibitory for IFN γ production from Th1 cells (Fig.15.5).

Role of Natural Killer (NK) cells in Cell Mediated Immunity

NK cells are also called as Large Granular Lymphocytes (LGL) as these resemble with lymphocytes except that they are larger and have numerous granules. These have surface molecules like C16 and CD56 but lack in CD3 molecule. NK cells have the ability of killing virus infected and malignant cells after activation. These are activated upon exposure to cytokines like IL-2 & IFN γ and then recognize the infected and malignant cells (Fig.15.6). The recognition is based on the absence of class I MHC molecule on the surface of infected cells in contrast to normal cells. Afterward, NK cells kill the target cell like CTL using perforin & graenzymes

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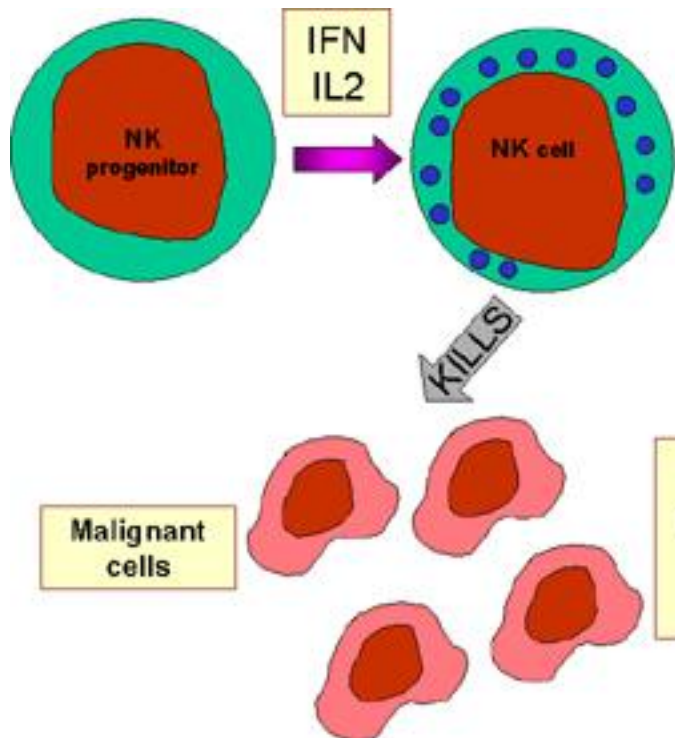


Figure 15.6: Activation & killing of malignant cells by NK cells

NK cells need cell to cell interaction for its effector function to kill malignant cells. NK cell cannot kill normal cells which are self cells due to normal expression of Class I MHC. However, the viral infected and malignant cells are recognized due to absence of Class I MHC (Fig.15.7). In this way viral infected & tumor cells are rejected

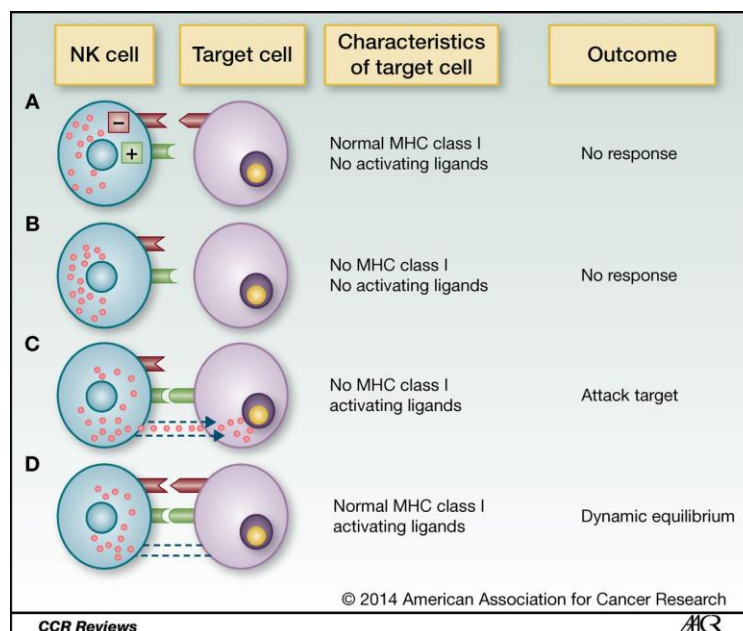


Figure 15.7: Functioning of NK cells based on characteristics of target cells

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Antibody dependent Cellular Toxicity by Killer Cells

Antibody dependent Cellular Cytotoxicity or (ADCC) is a kind of cellular killing which is based on antibody and killer (K) cells. K cells are not morphologically distinct type of cells. These are any cell which can mediate ADCC like NK cell, macrophages & PMNLs. Antibody act as to bring K cell close to target cell leading towards cell to cell interaction. K cells have Fc receptors (FcR) on their surface. For such kind of killing firstly antibodies like IgG which are specific to antigens of target cell bind to target cell. The coated target cell binds to Fc receptor for IgG present on NK, LAK cells and macrophages. Moreover, the killing of the target cell is being done through perforin/granzyme mediated mechanisms (Fig.15.8).

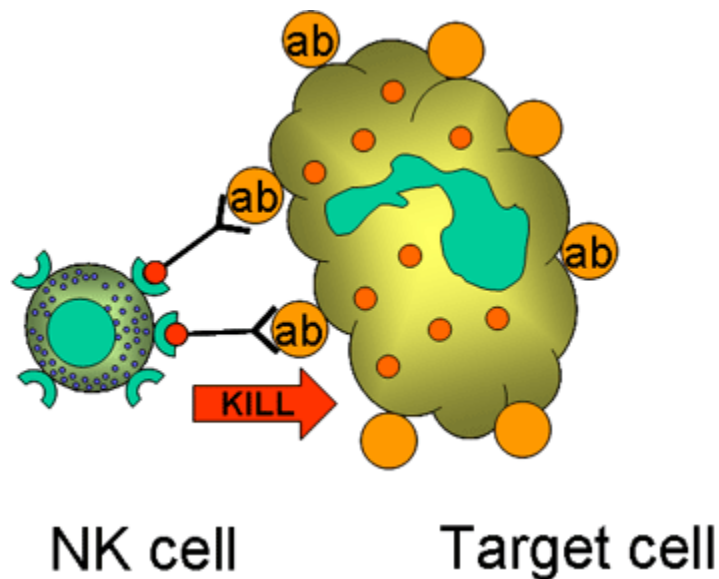


Figure 15.8: The process of Antibdy dependent cellular cytotoxicity (ADCC)

Furthermore, ADCC can also be mediated through IgE. For this the mast cells and eosinophils are involved which have Fc receptors for IgE. Firstly, IgE coat the parasite surface which then bring coated parasitic cell to mast cells. Mast cells or eosinophils get degranulated for killing of parasite (Fig.15.9)

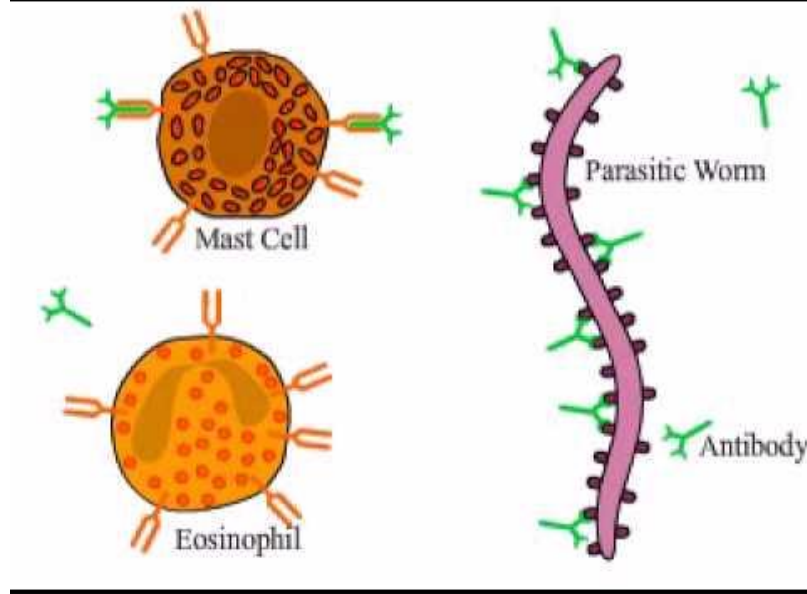


Figure 15.9: IgE mediated killing of parasite via ADCC

Lymphokine Activated Killer Cells in Cell Mediated Immunity

Lymphokine activated killer (LAK) cells are NK cells which are differentiated by continuous exposure to IL-2 and interferon (Fig.15.10). Upon activation, LAK cells kill the transformed and malignant cells. This form killing process is selective for sparing normal cells but need cell to cell interaction

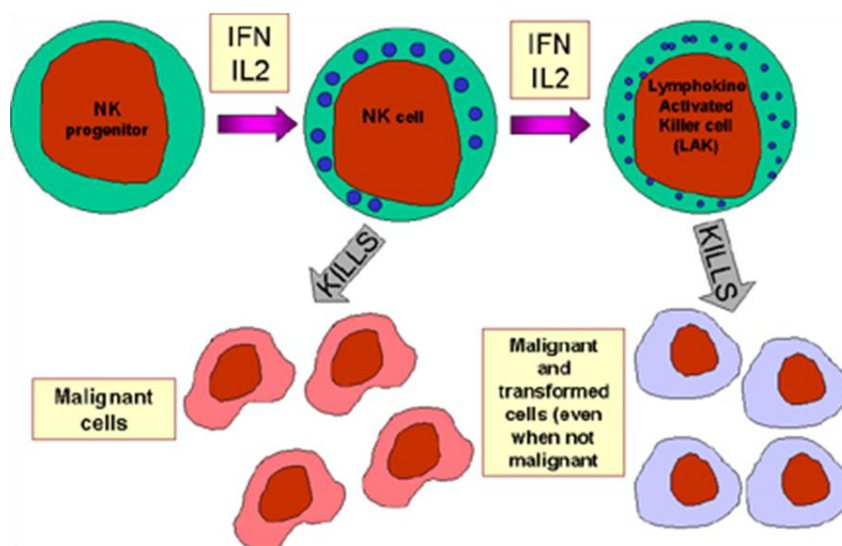


Figure 15.10: Activation of LAK cells

The killing is executed as tumor cells lack class I MHC, LAK cells are specific for tumor cells. LAK cells respond to those tumor cells which are resistant to NK cells killing. LAK cell therapy is used for treating various tumors.

